Final Exam Introduction to Econometrics 2020 (EBB828A05) June 26th

1. (50 points) Consider a sample of 4291 sampling units (areas) in Sub-Saharan Africa. You have information on the share of households connected to electricity ($connected_i$, measured as a fraction between 0 and 1), reliability of electricity connection in the area (rel_i), share of skilled workers in the area ($skilled_i$), share of individuals who hold a full time cash paying job ($jobcash_i$), and dummy variables indicating

   - presence of a bank in the area ($bank_i$)
   - presence of a market in the area ($market_i$)
   - average wealth of a household in the area ($high_i$, $medium_i$, and low_i)
   - whether the area is an urban area ($urban_i$)

   Consider the following OLS regression for an area i:

   $$connected_i = \beta_1 + \beta_2 rel_i + \beta_3 skilled_i + \beta_4 jobcash_i + \beta_5 bank_i + \beta_6 market_i + \beta_7 high_i + \beta_8 medium_i + \beta_9 urban_i + \varepsilon_i \quad (1)$$

   The results of regression (1) are presented in Table 1 (cf. column 'equation 1').

   (a) Provide a careful economic interpretation of the estimated coefficient on $bank_i$.

   > **Solution:** Keeping other factors constant, areas with a bank have 0.057 higher electricity connection rate than areas without a bank (the reference group).

   (b) Table 1 also presents the results of a regression model without the wealth dummies (cf. column 'equation 2'). Check by means of a F test whether the wealth dummies are jointly significant. Please mention all steps of the testing procedure ($\alpha = 0.05$).

   > **Solution:** Steps F-test
   >
   > - Formulate H_0 and H_1
   > $H_0 : \beta_7 = 0, \beta_8 = 0$, $H_1 : H_0$ not true
   > - Test statistic
   > $$F(\#r, n-K) = \frac{(SSR_r - SSR_u)/\#r}{SSR_u/(n-K)}$$

This test statistic follows a student F distribution with $\#r$ numerator degrees of freedom and $n-K$ denominator degrees of freedom.

In this case, $n = 4291, K = 9, \#r = 2$,

- Compute the value of the F statistic. Notice that $SSR_r = 257.2051, SSR_u = 256.1648$

$$F(2, 4291-9) = \frac{(257.2051 - 256.1648)/2}{256.1648/(4291-9)} = 8.7$$

- Look up critical value ($\alpha = 0.05, \#r = 2, n-K = 4291-9$): 3.000
- Since $8.7 > 3.000$. So the wealth dummies $high_i$ and $medium_i$ are jointly statistically significant at the 5 % level.

(c) Now, we consider the possibility that the β coefficients differ across subgroups of urban ($urban_i = 1$) and rural ($urban_i = 0$) areas. We have estimated the following model for both subgroups (cf. columns (`urban`) and (`rural`) in Table 1):

$$connected_i = \beta_1^k + \beta_2^k rel_i + \beta_3^k skilled_i + \beta_4^k jobcash_i + \beta_5^k bank_i + \beta_6^k market_i + \beta_7^k high_i + \beta_8^k medium_i + \varepsilon_i \tag{2}$$

where k= urban, rural. Test the null hypothesis that all *slope* coefficients of model (2) are the same across the subgroups of "urban" and "rural" ($\alpha = 0.05$).

Solution: Steps F test

- Formulate H_0

$$H_0 : \beta_i^{rural} = \beta_i^{urban}, i = 2, \ldots, 8$$

Apart from the intercept term, the coefficients of the "rural equation" are the same as those of the "urban equation".

Number of restrictions imposed by H_0: $\#r = 7$ (2*8 parameters unrestricted model and 9 parameters in restricted model, also 16-9=7 restrictions)

- Computation F statistic

$$F(\#r, n-K) = \frac{(SSR_r - SSR_u)/\#r}{SSR_u/(n-K)}$$

where K = number of regression parameters in unrestricted model

Unrestricted model (separate equations for urban=1 and urban=0)

Number of regression parameters in unrestricted model: $2 \times 8 = 16$

SSR unrestricted model: $SSR_u = SSR^{urban} + SSR^{rural} = 95.2584 + 152.3976 = 247.656$

SSR restricted model: $SSR_r = 256.1648$ (see column 'equation 1 table 1)

So, The value of the F-statistic equal to:

$$F(7, 4291 - 16) = \frac{(256.1648 - 247.656)/7}{247.656/(4291 - 16)} = 20.98$$

- Look up critical value: $c = 2.010$
- $20.98 > 2.010$, we reject the H_0. We conclude that (at least one of the) slope coefficients is different across the subgroups of urban and rural areas.

(d) Now we extend equation (1) as follows:

$$connected_i = \beta_1 + \beta_2 rel_i + \beta_3 skilled_i + \beta_4 skilled_i^2 + \beta_5 jobcash_i + \beta_6 bank_i + \beta_7 market_i + \beta_8 high_i + \beta_9 medium_i + \beta_{10} urban_i + \beta_{11} urban_i * skilled_i + \beta_{12} urban_i * skilled_i^2 + \varepsilon_i \tag{3}$$

The results of this regression are also presented in Table 1 (cf. column 'interactions')

What is the marginal effect of an increase in the share of skilled individuals in an area on the connection rate in an urban area ($urban_i = 1$) with 12% of skilled workers ($skilled_i = 0.12$)?

Solution: For urban areas the marginal effect at $skilled = 0.12$: $b_{skilled} + 2 \cdot b_{skilled^2} \cdot 0.12 + b_{urban*skilled} + 2 \cdot b_{urban*skilled^2} \cdot 0.12 = 0.483 + 2 * (-0.0776) * 0.12 - 0.158 + 2 * (-0.0581) * 0.12 = 0.29$

(e) You observe that the variable rel_i can be endogenous in this model. You also know that in Sub Saharan Africa reliability of electricity supplied depends on the density of transmission network in the area. You decide to use the density of transmission network ($trans_dens_i$) as an instrument for reliability (rel_i). In other words, we consider the following reduced form equation for rel_i ("first stage equation"):

$$rel_i = \gamma_1 + \gamma_2 trans_dens_i + \gamma_3 skilled_i + \gamma_4 skilled_i^2 + \gamma_5 jobcash_i + \gamma_6 bank_i + \gamma_7 market_i + \gamma_8 high_i + \gamma_9 medium_i + \gamma_{10} urban_i + \gamma_{11} urban_i \cdot skilled_i + \gamma_{12} urban_i \cdot skilled_i^2 + \varepsilon_i \tag{4}$$

The parameters of equation (3) are estimated by means of 2SLS and GMM (cf. columns '2sls' and 'gmm' of Table 1). Table 1 also presents the estimation results of equation (4) (see column 'first stage').

What conditions regarding $trans_dens_i$ need to be satisfied for the 2SLS estimator to yield consistent estimators of the β parameters in equation (3)? Do you think that these conditions are satisfied in this application?

Solution: Relevance and validity

(f) Use the information presented in Table 1 to test the hypothesis that in equation (3) the right hand side variable rel_i is exogenous ($\alpha = 0.05$).

Solution: Exogeneity test: $H_0 : E(rel_i u_i) = 0$
Hausman-Wu test of exogeneity can be carried by running the following auxiliary regression:

$$\begin{aligned} connected_i &= \beta_1 + \beta_2 rel_i + \beta_3 skilled_i + \beta_4 skilled_i^2 + \beta_5 jobcash_i + \beta_6 bank_i + \beta_7 market_i + \beta_8 high_i \\ &+ \beta_9 medium_i + \beta_{10} urban_i + \beta_{11} urban_i \cdot skilled_i + \beta_{12} urban_i \cdot skilled_i^2 \\ &+ \theta \hat{v}_i + \varepsilon_i \end{aligned} \tag{5}$$

where $\hat{v}_i$ denotes the residual of the first stage regression. The test statistic is equal to the t-value of the estimated parameter associated wit the first stage residual (see Table 1 column aux regression 1).

$$t_{\hat{v}} = \frac{-0.224}{0.0510} = -4.39$$

Since $|-4.39| > 1.96$ ($\alpha = 0.05$), the H_0 of exogeneity is rejected. We find evidence that the right hand side variable rel_i is endogenous.

(g) From Table 1 one can infer that 2SLS and GMM yield the same parameter estimates. Can you provide an explanation for this phenomenon?

Solution: The model presented in part e is 'just identified': $K = L$, number of instruments (K) equal to number of regressors in model (3). In the 'just identified case the choice of the weighting matrix is not relevant, The 2SLS and GMM estimator are both equal to:

$$\hat{\boldsymbol{\beta}}_{2sls} = \hat{\boldsymbol{\beta}}_{gmm} = (\mathbf{X}'\mathbf{Z})^{-1}\mathbf{X}'\mathbf{y}$$

2. (15 points) Please provide a short answer to the following questions.

(a) WLS is preferred to OLS when an important variable has been omitted from the model. True, false or uncertain? Explain.

Solution: False. The unbiasedness of WLS and OLS hinges crucially on strict exogeneity assumption. This assumption is often violated when an important variable is omitted. When strict exogeneity does not hold, both WLS and OLS are biased. Without specific information on how the omitted variable is correlated with the included explanatory variables, it is not possible to determine which estimator has a small bias. It is possible that WLS would have more bias than OLS or less bias. Because we cannot know, we should not claim to use WLS in order to solve "biases" associated with OLS.

The next two sub-questions refer to the following estimating scenario. Consider the following linear model to explain yearly beer consumption:

$$beer_{it} = \beta_0 + \beta_1 inc_{it} + \beta_2 price_{it} + \beta_3 educ_i + \beta_4 female_i + c_i + u_{it}$$

Where $E(u_{it}|inc_{it}, price_{it}, educ_i, female_i) = 0$ and $u_{it} \sim iid(0, \sigma_u^2)$. Subscripts i and t denote individual and year respectively. c_i denotes individual effect.

(b) Fixed effects estimator for β_3 is consistent. True, false or uncertain. Explain.

> **Solution:** False. β_3 is not identified with FE estimator.

(c) Since random effect (RE) estimator is efficient for β_1, we should use RE to estimate β_1. True, false or uncertain? Explain.

> **Solution:** Uncertain. It depends on whether c_i is correlated with x_{it}. If yes, then RE is not consistent for β_1 and thus it should not be used. If no, than RE is indeed a good choice.

3. (35 points) Let y be the dependent variable, $\mathbf{x}$ a $(K \times 1)$ vector of explanatory variables and s a binary group indicator. Define $\mathbf{u}_0(\mathbf{x}) = E(y|\mathbf{x}, s = 0)$ and $\mathbf{u}_1(\mathbf{x}) = E(y|\mathbf{x}, s = 1)$. In other words, it holds that

$$E(y|\mathbf{x}, s) = (1 - s)\mathbf{u}_0(\mathbf{x}) + s\mathbf{u}_1(\mathbf{x})$$

(a) Show that

$$E(y|s = 1) - E(y|s = 0) = (E(\mathbf{u}_1(\mathbf{x})|s = 1) - E(\mathbf{u}_0(\mathbf{x})|s = 1)) + (E(\mathbf{u}_0(\mathbf{x})|s = 1) - E(\mathbf{u}_0(\mathbf{x})|s = 0))$$

HINT: Use the law of iterated expectations

> **Solution:** According the law of the iterated expectations it holds that
>
> $$\begin{aligned} E(y|s) &= E(E(y|\mathbf{x}, s)|s) = \\ &= E((1 - s)\mathbf{u}_0(\mathbf{x}) + s\mathbf{u}_1(\mathbf{x})|s) \\ &= (1 - s)E(\mathbf{u}_0(\mathbf{x})|s) + sE(\mathbf{u}_1(\mathbf{x})|s) \text{ (linearity of conditional expectations)} \end{aligned}$$
>
> So
>
> $$\begin{aligned} E(y|s = 1) - E(y|s = 0) &= E(\mathbf{u}_1(\mathbf{x})|s = 1) - E(\mathbf{u}_0(\mathbf{x})|s = 0) \\ &= (E(\mathbf{u}_1(\mathbf{x})|s = 1) - E(\mathbf{u}_0(\mathbf{x})|s = 1)) + (E(\mathbf{u}_0(\mathbf{x})|s = 1) - E(\mathbf{u}_0(\mathbf{x})|s = 0)) \end{aligned}$$

(b) Suppose that both expectations are linear: $\mathbf{u}_0(\mathbf{x}) = \beta_0'\mathbf{x}$ and $\mathbf{u}_1(\mathbf{x}) = \beta_1'\mathbf{x}$. Show that

$$E(y|s = 1) - E(y|s = 0) = (\beta_1 - \beta_0)'E(\mathbf{x}|s = 1) + \beta_0'(E(\mathbf{x}|s = 1) - E(\mathbf{x}|s = 0))$$

> **Solution:**
>
> $$\begin{aligned} E(y|s = 1) - E(y|s = 0) &= (E(\mathbf{u}_1(\mathbf{x})|s = 1) - E(\mathbf{u}_0(\mathbf{x})|s = 1)) + (E(\mathbf{u}_0(\mathbf{x})|s = 1) - E(\mathbf{u}_0(\mathbf{x})|s = 0)) \\ &= (E(\beta_1'\mathbf{x}|s = 1) - E(\beta_0'\mathbf{x}|s = 1)) + (E(\beta_0'\mathbf{x}|s = 1) - E(\beta_0'\mathbf{x}|s = 0)) \\ &= (\beta_1 - \beta_0)'(E(\mathbf{x}|s = 1) + \beta_0'(E(\mathbf{x}|s = 1) - E(\mathbf{x}|s = 0)) \end{aligned}$$

Table 1: Estimates results for question 1

VARIABLES	equation 1	equation 2	urban	rural	interactions	first stage	2sls	gmm	aux. regression 1
rel	0.204	0.204	0.117	0.265	0.196		0.445	0.445	0.445
	(0.0104)	(0.0104)	(0.0126)	(0.0164)	(0.0104)		(0.0520)	(0.0405)	(0.0495)
skilled	0.283	0.297	0.204	0.380	0.483	0.749	0.395	0.395	0.395
	(0.0150)	(0.0146)	(0.0162)	(0.0272)	(0.0630)	(0.0996)	(0.0838)	(0.0888)	(0.0799)
skilledsq					-0.0776	-0.671	0.00665	0.00665	0.00665
					-0.00742	(0.117)	(0.0924)	(0.0949)	(0.0880)
jobcash	0.0583	0.0518	0.0307	0.0951	0.0611	-0.0304	0.0598	0.0598	0.0598
	(0.0147)	(0.0146)	(0.0171)	(0.0240)	(0.0146)	(0.0242)	(0.0174)	(0.0192)	(0.0165)
bank	0.0574	0.0587	0.0481	0.0905	0.0555	0.0911	0.0308	0.0308	0.0308
	(0.00921)	(0.00922)	(0.00913)	(0.0199)	(0.00914)	(0.0158)	(0.0117)	(0.0110)	(0.0112)
market	0.0242	0.0245	-0.0115	0.0281	0.0134	0.0445	0.0163	0.0163	0.0163
	(0.00882)	(0.00883)	(0.0109)	(0.0139)	(0.00884)	(0.0147)	(0.0106)	(0.0106)	(0.0101)
high	0.0478		0.0400	0.0732	0.0551	0.0212	0.0447	0.0447	0.0447
	(0.0118)		(0.0133)	(0.0200)	(0.0118)	(0.0192)	(0.0139)	(0.0138)	(0.0133)
medium	0.0258		0.0342	0.0173	0.0260	0.00808	0.0212	0.0212	0.0212
	(0.00879)		(0.0101)	(0.0146)	(0.00872)	(0.0147)	(0.0106)	(0.0113)	(0.0101)
urban	0.140	0.139			0.208	0.295	0.149	0.149	0.149
	(0.00872)	(0.00873)			(0.0178)	(0.0293)	(0.0270)	(0.0254)	(0.0258)
urban_skilled					-0.158	-0.661	-0.0147	-0.0147	-0.0147
					-0.00888	(0.146)	(0.112)	(0.109)	(0.107)
urban_skilledsq					-0.0581	0.585	-0.195	-0.195	-0.195
					-0.00958	(0.155)	(0.118)	(0.111)	(0.113)
trans_dens						5.996			
						(0.432)			
vhat									-0.224
									(0.0510)
Constant	0.379	0.399	0.654	0.307	0.344	0.159	0.256	0.256	0.256
	(0.0116)	(0.0101)	(0.0162)	(0.0174)	(0.0131)	(0.0214)	(0.0193)	(0.0180)	(0.0184)
Observations	4,291	4,291	2,340	1,951	4,291	3,070	3,070	3,070	3,070
R-squared	0.363	0.360	0.135	0.284	0.373	0.205	0.387	0.387	0.446
SSR	256.1648	257.2051	95.2584	152.3976	251.9930	359.9133	187.6314	187.6314	169.5078
F test exclusions restrictions						45.678			0.54

Standard errors in parentheses; vhat denotes the residuals from the first stage